ALA-TOO INTERNATIONAL UNIVERSITY

BUSINESS ADMINISTRATION SOFTWARE

Spring Semester 2025-2026

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Without data,

you're just another person with an opinion.

– W. Edwards Deming

ANALYTICS & DASHBOARDS

COURSE PROGRESSION

ERP → CRM → ANALYTICS

ERP records operations

CRM records customers

Analytics transforms system data into decision control

This week: control through measurement

THE DATA PROBLEM

OVERLOAD AND COMPLEXITY

ERP and CRM systems generate millions of transactions

Managers CANNOT interpret raw operational logs

They need structured signals

WHAT IS ANALYTICS?

FORMAL DEFINITION

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FORMAL DEFINITION

Analytics is the structured transformation of operational data into measurable indicators that reduce uncertainty and support decision-making.

WHAT IS KPI?

FORMAL DEFINITION

WHAT IS KPI?

FORMAL DEFINITION

A Key Performance Indicator (KPI) is a quantifiable metric that reflects how effectively an organization, department, or process is achieving a defined business objective. It is directly linked to strategy, has clearly defined measurement criteria, and includes performance thresholds that trigger evaluation or corrective action.

KPI

ANALOGY

When you drive, you do not monitor every engine component. You look at indicators such as speed, fuel level, or engine temperature.

- ➢ If the temperature rises above a safe limit, you take action.
- ➢ If fuel is low, you refuel.

Similarly, managers do not monitor every transaction. They monitor KPIs such as revenue, conversion rate, or cost variance. When a KPI crosses a defined threshold, action is required. A KPI is not just a number. It is a signal that guides decisions.

LEADING KPIs

PREDICTING FUTURE PERFORMANCE

LEADING KPIs

PREDICTING FUTURE PERFORMANCE

Leading KPIs measure activities and trends that influence future outcomes.They act as early warning signals before financial results are visible. Leading KPIs are proactive.They allow intervention before damage occurs.

Examples:

- ➢ Conversion rate (declining conversion predicts revenue drop)
- ➢ Pipeline velocity (slower movement predicts delayed income)
- ➢ Customer engagement rate (low engagement predicts churn)
- ➢ Sales cycle length (increasing duration predicts slowdown)

LAGGING KPIs

CONFIRMING PAST RESULTS

LAGGING KPIs

CONFIRMING PAST RESULTS

Lagging KPIs measure outcomes that have already occurred. They confirm whether business objectives were achieved. Lagging KPIs are reactive. They show results, but not necessarily the cause.

Examples:

- Revenue (total money generated from sales before expenses)
- Gross profit (revenue minus the direct cost of producing goods or services)
- Net income (final profit after all expenses, taxes, and costs are deducted)
- Quarterly growth (percentage increase or decrease in performance compared to the previous quarter)

KPI TYPES

LEADING VS LAGGING

Leading KPIs predict outcomes

Lagging KPIs confirm past results

ANALOGY

VITAL SIGNS MODEL

Doctors monitor heart rate, not individual cells.

Managers monitor KPIs,
NOT transactions.
(Dashboards are business vital signs)

DATA TRANSFORMATION PIPELINE

FROM TRANSACTION TO ACTION

Business systems generate raw transactions such as orders, payments, invoices, and customer interactions

Transaction

Individual recorded event (one customer order worth $200)

→ Aggregation → Metric → KPI → Threshold → Decision

DATA TRANSFORMATION PIPELINE

FROM TRANSACTION TO ACTION

Aggregation

Combining multiple transactions into summarized data (total daily sales)

Metric

Calculated measurement derived from aggregated data (average order value)

KPI

Strategic metric linked to a business objective (monthly revenue target achievement)

DATA TRANSFORMATION PIPELINE

FROM TRANSACTION TO ACTION

Threshold

Defined performance limit that separates acceptable from unacceptable results (profit margin below 15%)

Decision

Management action taken when a KPI crosses its threshold (reduce costs, adjust pricing, review sales process)

RAW DATA VS METRIC

ADDING INTERPRETATION

Data by itself does not explain performance.Metrics give data meaning. Metrics transform events into performance signals

Raw Data: 10,000 orders (this tell us that the company processed 10,000 orders. That’s it.)

Is that good or bad?

Were they delivered on time?

Were they profitable?

Did customers wait too long?

RAW DATA VS METRIC

ADDING INTERPRETATION

Metric: Average fulfillment time = 3.2 days

Fulfillment time measures: the time between when a customer places an order and when the order is delivered

With this metric we are measuring performance? How?

RAW DATA VS METRIC

ADDING INTERPRETATION

3.2 days tells us:

On average, customers waited 3.2 days to receive their order

We can compare it to:

- ➢ Last week (better of worse)
- ➢ A company target (for example 2 days)
- ➢ Competitors (example - 4 days)

Now we can evaluate performance

METRIC
EXAMPLE

For each order, the ERP system stores (raw data):

- Order Date
- Delivery Date

Order ID	Order Date	Delivery Date
1001	Jan 1 2026	Jan 3 2026
1002	Jan 1 2026	Jan 4 2026
1003	Jan 2 2026	Jan 5 2026

METRIC

EXAMPLE

Calculate fulfillment time per order

Order 1001 → 2 days

Order 1002 → 3 days

Order 1003 → 3 days

Calculate the average

Average Fulfillment Time = (2 + 3 + 3) ÷ 3 = 2.67 days

KEY PERFORMANCE INDICATORS

STRUCTURE AND PURPOSE

A KPI key prformance indicators:

- ➢ Links to a business objective (directly supports a specific strategic goal such as growth, profitability, or efficiency)
- ➢ Has ownership (assigned to a responsible person or department accountable for its performance)
- ➢ Has measurable limits (includes defined targets or thresholds that distinguish acceptable from unacceptable performance)
- ➢ Triggers corrective action (requires a response or intervention when performance falls outside defined limits)

KPI LIFECYCLE

DEFINE, MONITOR, REVISE

KPIs must evolve with strategy

As business priorities change, for example from growth to profitability, or from expansion to efficiency performance indicators must also change to reflect new objectives

Static KPIs eventually mislead

If organizations continue measuring outdated targets, they may optimize the wrong behavior, ignore emerging risks, or fail to respond to market shifts

KPI MANIPULATION RISK

BEHAVIORAL DISTORTION

If sales performance is measured only by call volume,
sales representatives will increase the number of calls made,
but the quality of conversations and customer qualification may decline

When a KPI rewards quantity instead of outcome, behavior adapts to maximize the metric rather than the objective

Wrong KPI → wrong behavior

ERP ANALYTICS EXAMPLE

INVENTORY TURNOVER

Inventory Turnover = Cost of Goods Sold / Average Inventory

Declining turnover signals:

- **Cash locked in stock** (capital tied up in unsold inventory instead of being used elsewhere in the business)
- **Storage cost increase** (higher warehousing, insurance, and handling expenses due to excess stock)
- **Operational inefficiency** (slow-moving products indicate weak demand forecasting or purchasing problems)

ERP COST CONTROL

VARIANCE DETECTION

Cost Variance = Actual Cost − Planned Cost

Threshold breach triggers:

- ➢ Procurement review (evaluate purchasing decisions, order quantities, and supplier selection to identify cost inefficiencies)
- ➢ Supplier renegotiation (reassess pricing terms, payment conditions, or delivery agreements to reduce expenses)
- ➢ Efficiency investigation (analyze internal production or operational processes to identify waste, delays, or resource misallocation)

CRM ANALYTICS EXAMPLE

PIPELINE VELOCITY

Pipeline Velocity =(Opportunities × Win Rate × Deal Size) / Sales Cycle Length

Declining velocity predicts revenue slowdown

This metric estimates how quickly revenue is being generated from the sales pipeline.

- ➢ More opportunities increase potential revenue.
- ➢ Higher win rate improves conversion efficiency
- ➢ Larger deal size increases revenue impact
- ➢ Longer sales cycle slows revenue realization.

If pipeline velocity declines, it signals that future revenue growth may slow, even if current revenue still appears stable.

AMAZON

OPERATIONAL CONTROL SYSTEM

Amazon monitors:

- ➢ Late shipment percentage
- ➢ Fulfillment time
- ➢ Inventory aging

Micro signals prevent macro losses

AMAZON

OPERATIONAL CONTROL SYSTEM

Late shipment percentage (percentage of orders delivered after the promised delivery date)

Fulfillment time (average time between order placement and successful delivery to the customer)

Inventory aging (length of time products remain in storage before being sold)

NETFLIX

CONTENT PERFORMANCE METRICS

Netflix tracks:

- Completion rate
- Engagement time
- Subscriber churn

Analytics determines content investment.

NETFLIX

CONTENT PERFORMANCE METRICS

Completion rate (percentage of viewers who watch content from start to finish)

Engagement time (average amount of time users actively spend interacting with the platform or content)

Subscriber churn (percentage of subscribers who cancel their subscription within a given period)

ZILLOW

MODEL LIMITATION CASE

Predictive pricing failed due to:

- ➢ Static assumptions
- ➢ Market volatility
- ➢ Delayed feedback loops

Result: $500M+ loss.

ZILLOW

MODEL LIMITATION CASE

Static assumptions (models built on fixed expectations that do not adjust when market conditions change)

Market volatility (rapid and unpredictable changes in prices, demand, or economic conditions)

Delayed feedback loops (slow detection of performance problems due to lag between data collection, analysis, and corrective action)

REPORTS VS DASHBOARDS

STATIC VS DYNAMIC

Reports explain history

They summarize past performance and answer questions such as "What happened last month?" or "How did we perform last quarter?"

Dashboards enable intervention

They monitor current performance in near real time and highlight when a KPI crosses a defined threshold, allowing managers to take corrective action immediately.

Report → understanding the past

Dashboard → acting in the present

MULTIPLE VERSIONS OF TRUTH

METRIC CONSISTENCY RISK

Sales revenue ≠ Finance revenue

Sales may record revenue when a deal is closed in the CRM system, while Finance records revenue only after the invoice is issued or payment is received in the ERP system

Different definitions create confusion

When departments calculate metrics using different rules, dashboards show conflicting numbers and management loses trust in the data

IT must enforce unified metrics

Data definitions, calculation logic, and reporting rules must be standardized across systems to ensure one version of the truth

DASHBOARD DESIGN RULE

ROLE-SPECIFIC CONTROL

One dashboard must serve:

- One role
- One decision scope
- One timeframe

IT RESPONSIBILITY

DATA MODELING AND GOVERNANCE

IT professionals:

- ➢ Define data structures
- ➢ Maintain ETL logic
- ➢ Enforce KPI definitions
- ➢ Prevent duplication

Analytics integrity is technical work.

P001 - KPI CALCULATION ENGINE

FROM DATA TO METRICS

Input:

- ➢ Leads
- ➢ Conversion rate
- ➢ Average deal value
- ➢ Marketing cost
- ➢ Operational cost

Output:

- ➢ Revenue
- ➢ Gross Profit
- ➢ CAC
- ➢ Conversion classification

P002 – DECISION ENGINE

FROM KPI TO ACTION

Based on KPI classification:

- Critical → Review sales process
- Negative profit → Reduce costs
- Strong performance → Scale marketing

Output exactly one management action.

P003 – TREND ANALYSIS ENGINE

COMPARING TWO PERIODS

The program must compare performance between two time periods:

- ➢ Previous quarter
- ➢ Current quarter

The system receives the following inputs for both periods:

- ➢ Revenue
- ➢ Gross Profit
- ➢ Conversion Rate
- ➢ Number of Leads

P003 – TREND ANALYSIS ENGINE

TASK REQUIREMENT

The program must detect performance contradictions that indicate hidden risk.

Specifically, detect the following situations:

1. Revenue increased BUT Gross Profit decreased
2. Number of Leads increased BUT Conversion Rate decreased

If either condition is true, the system must output: “Hidden Risk Detected”

If neither condition is true, the system must output: “Performance Stable”

P003 – TREND ANALYSIS ENGINE

WHY THIS MATTERS

Revenue alone does not guarantee healthy performance

Growth without profitability or efficiency can signal structural problems

Analytics must detect contradictions, not just growth

TREND INTERPRETATION

DIRECTION OVER SNAPSHOT

Static numbers can mislead

Revenue of $1 million may look strong, but if it has declined for three consecutive months, it signals emerging risk

Direction determines strategy

Trends over time → increasing, stable, or declining → guide long-term decisions and resource allocation

WHAT IF

MISLEADING DASHBOARD

The executive dashboard shows:

- ➢ Revenue appears stable
- ➢ Lead volume is increasing
- ➢ Conversion rate is declining
- ➢ Profit margin is shrinking

At first glance, growth seems healthy. However, deeper indicators reveal weakening efficiency and declining profitability. The dashboard highlights activity, but hides structural risk.

WHAT IF

MISLEADING DASHBOARD

Same case but in other words

Revenue looks stable but marketing reports more leads.

But conversion efficiency is dropping and profit margins are tightening.

The business appears stable, yet the foundation is weakening.

BUSINESS CONSEQUENCE

WRONG DECISION IMPACT

Management increases marketing spend based on stable revenue and rising lead volume

However, the underlying issues → declining conversion and shrinking margins → remain unresolved

More leads enter an inefficient sales process.

- ➢ Customer acquisition costs increase
- ➢ Operational pressure rises
- ➢ Profitability declines

An analytics interpretation failure becomes a financial performance failure

ANALYTICS MINDSET

REDUCING UNCERTAINTY

Analytics does not replace managerial judgment, and it does not automate thinking

Its purpose is to identify patterns, detect deviations, and highlight emerging risks before they become visible in financial results

By transforming system data into structured signals, analytics reduces decision uncertainty and supports more informed, evidence-based actions

FINAL TAKEAWAY

HAAAAA

SYSTEM → SIGNAL → CONTROL

ERP records operations

CRM records customers

Analytics transforms signals into control

Dashboards shape behavior.

Poor dashboards shape mistakes.

NEXT WEEK PREVIEW

WHAT'S COMING

Designing KPIs and decision-safe dashboards